

EXAM 4

Crystal Bowers

1) $T = 625 \text{ K}$

$t = 500 \text{ days}$

$PBR = 1.56$

$t^0 = 600 \mu\text{m}$

$\rho_{Zr} = 6.5 \text{ g/cm}^3$

a) oxide thickness

$\rho_{ZrO_2} = 5.68 \text{ g/cm}^3$

b) H pickup $f = 14\%$

what's the wt ppm of H in cladding?

a) $t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right) = 133 \text{ days}$

$\uparrow$
625K

$\delta^*(\mu\text{m}) = 5.1 \exp\left(\frac{-550}{T}\right) = 2.115 \mu\text{m}$

$\uparrow$
625K

$t > t^*$
so linear
approx.

($\mu\text{m/d}$)

$K_L = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right) = 0.0154 \frac{\mu\text{m}}{d}$

$\uparrow$
625K

$\delta(\mu\text{m}) = \delta^* + K_L(t - t^*)$

$= 2.115 \mu\text{m} + 0.0154 \frac{\mu\text{m}}{d} (500 \text{ days} - 133 \text{ days}) = 7.77 \mu\text{m}$

1b)

$$C_H = \frac{\check{\check{2f}} \check{\delta} \check{P_{oxide}} \check{f_{ZrO_2}^0} \check{\frac{M_H}{M_O}}}{\left(\check{t} - \check{\frac{\delta}{PBR}} \right) \check{P_{metal}}} \times 10^6$$

initial
clad
thick

~~Remember to add the~~

$$f_{ZrO_2}^0 = \frac{16 \times 2}{16 \times 2 + 91} = 0.26$$

$$C_H = \frac{(2)(0.14)(5.68 \frac{g}{cm^3})(0.26)(\frac{1}{16})^{(7.77 \mu m)}}{(600 \mu m - \frac{7.77 \mu m}{1.56})(6.5 \frac{g}{cm^3})} \times 10^6$$

DON'T
forget
↓

$$C_H = \frac{51.92}{\cancel{51.92}} \text{ wt ppm}$$

not given

C_H^0 to add
to.

2) Rate limiting step is diffusion of oxygen through the ZrO_2 oxide.

3) ~~low~~ A hydride periphery rim forms in the cladding and possibly blisters with deep depth because of the Soret effect, ~~and H_2 migration~~
~~and H_2 migration~~, and hydrogen migrates to areas with tensile stress. Soret effect means hydrogen migrates to areas with low temperature. Impacts of hydrides include reduction of fracture toughness and hydrogen embrittlement which can eventually lead to cladding failure depending on the conditions present.

4) RIA is a reactivity insertion accident in which there's a large and rapid reactivity insertion by a control rod ejection (PWRs) or control rod drop (BWRs). In a RIA, temperature and power spike. Pellets swell and gas bubble pressures increase. Failure by PCMI (pellet cladding mechanical interaction) or departure from nucleate boiling can occur. The control rod ejection in a PWR is caused by the failure of the control rod drive mechanism or housing failing and coolant pressure leading to ejection. The drop in a BWR can occur due to failure of the control rod ~~drive~~ housing too. It's a Design basis accident.

5) LOCA (loss of coolant accident) is a design basis accident in which a coolant pipe breaks leading to loss of coolant. LOCA is different from a RIA because different methods of initiation as described and it's slower.

LOCA impacts include temperature increase from decay heat and lack of cooling. Coolant system pressure drop combined with internal fuel rod pressure increasing leading to it ballooning or bursting.

6) Burnup impacts the type of failure and the failure probability during an accident by at ~~a~~ a higher burnup, the cladding has ~~been~~ undergone sufficient ~~the~~ Hydrogen embrittlement leading to an increase in the failure probability of cladding during an accident. ^{The cladding} ~~it~~ would have undergone a lot of stress leading to PCMI in the case of a reactivity insertion accident. Irradiation reduces the melting temperature and thermal conductivity of the fuel. The heat capacity of the fresh fuel system is more sensitive to changes, but an irradiated fuel system at higher burnup will be more likely to fail during an accident as it's experienced more stresses and embrittlement/corrosion.

7) 4 pathways to make fuel / cladding more tolerant are - improved ~~the~~ reaction kinetics with steam, improved cladding properties, improved fuel properties, and enhanced fission product retention.

8) One near-term ATF concept is improving Zr cladding using metallic, MAX, or ceramic nitride coatings. This is of interest to improve corrosion resistance and safety / performance benefits. It can be incorporated in typical LWRs in approximately 10 years with licensing efforts.

a) Zirconium cladding exposed to high temperature steam leads to oxidation. At $\sim 863^\circ\text{C}$ there's a $\alpha\text{-Zr}$ to $\beta\text{-Zr}$ phase transition. With high oxygen content, ~~the~~ stable oxygen ^{present} α can form with the $\beta\text{-Zr}$ leading to more brittle behavior. Above $\sim 1200^\circ\text{C}$, the cladding is entirely oxidized. Thick oxide layer that flakes at $\sim 1000^\circ\text{C}$.

10) ^{List} Limiting phenomena in LWR operations:

#1) Cladding wear

#2) PCMI (pellet ~~on~~ cladding mechanical interaction)

#3) cladding elongation and assembly bow

11) Fuel performance and safety impacts of CRUD include the safety risk of increased dose to radiation workers as CRUD can become activated as it's close to the fuel deposited on the cladding with radionuclides. It can then flake off and move to out-of-core components. Fuel performance wise, CRUD can trap boron in PWRs leading to CRUD power shifts. It reduces heat transfer.

12) Water chemistry controls that have been implemented in LWRs include Zinc injection in both PWRs/BWRs for radiation control. Boron is used as a poison / chemical shim to control reactivity so this is paired with LiOH control to maintain a constant pH so both can be reduced over time. This is performed to eliminate CRUD failures as a constant pH of > 6.9 has been determined to do this. H injection in BWRs is used to address IG SCC of core internals.

13) Microstructural phenomena unique to MOX:

~~Equiaxed grains~~

Fuel restructuring with four regions of as-sintered region, equiaxed ^{grain} region, columnar grains, and central void.

Central void forms from the accumulation of pores/voids at the center of

the fuel. The pores have gas with a higher temperature gradient leading to a condensation/evaporation mechanism. As these pores move they destroy fuel microstructure leaving

dense pack columnar grains. At lower temperatures, there are equiaxed grains that had growth compared to unirradiated MOX. ~~and~~ At

the periphery there's the as-sintered region without these changes due to lower temperatures. For MOX, the

center is Pu enriched and Pu depleted at the periphery columnar grains.

13 cont.) There's also oxygen diffusion
until Mo begins to oxidize then
it approaches stoichiometry
of 2.0.